

XIANGLIN HUANG

Ph.D., Mechanical Engineering · Impact Dynamics & Data-Driven Constitutive Modeling · Lithium-Ion Battery Impact Safety & PHM

Postdoctoral Researcher, InnoCORE PRISM-AI Center, KAIST · Visiting Research Professor, Dongguk University–Seoul

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RESEARCH SUMMARY

I develop data-driven methods for determining and extrapolating the dynamic flow stress of metals under impact loading, and lead an independent research stream on the impact safety and prognostics & health management (PHM) of lithium-ion batteries. A four-paper series in the International Journal of Impact Engineering (2023–2027) establishes a complete framework — combining SVD/CP tensor decomposition with artificial neural networks — for verifying when decoupled flow stress equations are legitimate, determining them from discrete SHPB data, and validly extrapolating them to very-high strain rates with Taylor-Hopkinson validation. In parallel, I supervise experimental and machine-learning research on lithium-ion battery failure detection, acoustic-emission monitoring, and multiphysics simulation of battery packs under dynamic loading.

Keywords: dynamic constitutive modeling · SHPB & Taylor-Hopkinson testing · tensor decomposition (SVD/CP) · machine learning for mechanics · crashworthiness & impact mechanics · lithium-ion battery impact safety · battery PHM · acoustic emission · multiphysics FEM

EDUCATION

University of Manchester (UoM) 08/2019 – 12/2024

Ph.D. in Mechanical Engineering · Department of Mechanical, Aerospace and Civil Engineering, School of Engineering

- Dissertation: **Determination of dynamic flow stress equation based on discrete material data**
- Supervisor: Prof. Qingming Li (Q.M. Li) — Qingming.li@manchester.ac.uk

Harbin Institute of Technology (HIT) 09/2015 – 07/2017

M.E. in Aeronautical and Astronautical Science and Technology · School of Astronautics

- Dissertation: **Experimental investigation on the impact resistance of polymer-aluminium layered plates**
- Supervisor: Prof. Wei Zhang — zhdawei@hit.edu.cn

Civil Aviation University of China (CAUC) 09/2010 – 07/2014

B.E. in Aircraft Manufacturing Engineering · College of Aeronautical Engineering

- Coursework: Mechanics of Materials, Aircraft Fault Diagnosis Technology, Engineering Materials, Aircraft Flight Mechanics, Aircraft Manufacturing Technology

ACADEMIC & PROFESSIONAL APPOINTMENTS

Postdoctoral Researcher, KAIST 10/2025 – Present

Supervisor: Prof. Seunghwa Ryu

- Prognostics and Health Management of Lithium-Ion Batteries Based on Neuromorphic Sensor and Artificial Intelligence Framework

Visiting Research Professor, Dongguk University 10/2025 – Present

Supervisor: Prof. Heung Soo Kim

- Prognostics and Health Management of Lithium-Ion Batteries Based on Neuromorphic Sensor and Artificial Intelligence Framework

Research Associate (Part-time), Zhejiang University, China

05/2025 – 08/2025

Supervisor: Prof. Guoxing Lu

- Correcting Load Ringing in Intermediate-Strain-Rate (VHS) Tensile Tests

Research Associate, University of Manchester

10/2020 – 04/2024

Supervisor: Prof. Q.M. Li

TEACHING EXPERIENCE

Instructor (Graduate Course), Dongguk University

03/2026 – 06/2026

Dept. of Mechanical, Robotics and Energy Engineering — “Machine Learning and Its Engineering Application”

- Sole instructor for a graduate-level course covering CNN, RNN, LSTM, and Transformer architectures, with emphasis on their application to mechanical and prognostics & health management (PHM) systems; students implemented models hands-on for data-driven diagnosis and prediction problems.
- Mentored a team of students competing in the 2nd KSPHM-KIMM Mechanical Data Challenge 2026, a national mechanical data-driven diagnostics competition (concluded June 25, 2026).

Teaching Assistant, University of Manchester

09/2019 – 12/2023

- Supported students completing the course Advanced Modelling & Simulation
- Supported students completing the course Composites & Polymers
- Demonstrated the operating procedure of column buckling experiments
- Instructed students completing ANSYS numerical simulation coursework

RESEARCH EXPERIENCE

Prognostics and Health Management of Lithium-Ion Batteries Based on Neuromorphic Sensor and Artificial Intelligence Framework

10/2025 – Present

- Conducting battery PHM research and leading a battery PHM research group

Multiscale and Multiphysics Coupled Failure Mechanisms of High Energy Density Batteries under Dynamic Loading

11/2024 – 04/2025

- Participated in drafting the proposal for a National Natural Science Foundation of China (NSFC) grant application, including the rationale, research content, objectives, and literature review. Published a paper in the *Journal of Energy Storage* as the corresponding author.

Damage Detection and Evaluation Model for Lithium-Ion Batteries under Dynamic Impact Based on Machine Learning

11/2023 – 06/2025

- Graduate Research Project (CAUC-funded): proposed the core concept of the research methodology and contributed to the writing and revision of a manuscript published in the *Journal of Power Sources* as the corresponding author.

Machine Learning–Based Ballistic Resistance Modelling of Metallic Targets

07/2023 – 03/2025

- Developed core research concepts and methodology; authored and revised manuscripts; published two articles as the corresponding author in *Thin-Walled Structures*.

Multiscale Modelling of Impact Behaviour in Angle-Interlock 3D Woven Composites

10/2023 – 11/2025

- Participated in constitutive modelling and simulation analysis; co-authored and published one article in the *International Journal of Mechanical Sciences* as a second author and another in *Composite Structures* as a second author.

Mechanical Response & Damage Mechanisms of Sandwich Structures

07/2015 – 06/2017

- Contributed to experimental design, data acquisition, and finite element simulations for an NSFC project at HIT; co-authored two articles published in the *International Journal of Impact Engineering*.

Application of Digital Image Correlation (DIC) in Sandbox Impact Experiments

05/2015 – 08/2015

- Designed and executed DIC-based penetration tests; analysed deformation fields and published findings at an American Institute of Physics (AIP) conference.

PUBLICATIONS (SCIE-INDEXED: 17, Q1: 16)

First authored	6
Corresponding author	5
Battery research related	5
Impact related	16
Composites	2
Sandwiches materials	3
Metallic Material Constitutive modelling	4
Composites Material Constitutive modelling	2
Foam material	1

First Authored

1. **X. Huang**, Q.M. Li, Tensor-decomposition-based inverse characterization of flow stress for very-high strain-rate impact. *International Journal of Impact Engineering*, 219 (2027) 105863 (JCR Q1, IF 6.7)
2. **X. Huang**, H.S. Kim, Advancing battery prognostics and health management: Challenges and future perspectives, *Journal of Mechanical Science and Technology*, 40(4) (2026) 2405–2415 (JCR Q3, IF 2.2)
3. **X. Huang**, Q.M. Li, Determination of dynamic flow stress equation based on discrete experimental data: Part 2 Dynamic flow stress depending on strain, strain-rate and temperature. *International Journal of Impact Engineering*, 206 (2025) 105432 (JCR Q1, IF 6.7)
4. **X. Huang**, Q.M. Li, Determination of dynamic flow stress equation based on discrete experimental data: Part 1 Methodology and the dependence of dynamic flow stress on strain-rate. *International Journal of Impact Engineering*, 206 (2025) 105403 (JCR Q1, IF 6.7)
5. **X. Huang**, Q.M. Li, The legitimacy of decoupled dynamic flow stress equations and their representation based on discrete experimental data. *International Journal of Impact Engineering* 173 (2023) 104453. (JCR Q1, IF 6.7)
6. **X. Huang**, W. Zhang, Y. Deng, X. Jiang, Experimental investigation on the ballistic resistance of polymer-aluminium layered plates. *International Journal of Impact Engineering* 113 (2018) 212–221. (JCR Q1, IF 6.7)

As Corresponding Author

(*Main contributions: Methodology, Conceptualization, Writing – original draft, review & editing)

7. Y. Deng, H. Zheng, T. Zhang*, **X. Huang***, Characterizing critical failure impact on LIB pack stability via multiphysics response, *Engineering Failure Analysis* 196 (2026) 111084. (JCR Q1, IF 7.0)
8. Y. Deng, J. Li, **X. Huang***, Deep learning-based real-time damage assessment of lithium-ion batteries under dynamic impact, *Journal of Power Sources*, 662 (2026) 238737. (JCR Q1, IF 8.4)
9. Y. Deng, H. Zheng, **X. Huang***, Failure modes and failure energy threshold of lithium-ion batteries under extreme impact, *Journal of Energy Storage*, 150 (2026) 120037. (JCR Q1, IF 10.7)
10. Y. Deng, Y. Lv, X. Yang, C. Du, **X. Huang***, Determination of residual velocity model of finite thickness material based on Artificial Neural Network. *Thin-Walled Structures*, 216 Part B (2025) 113685. (JCR Q1, IF 8.3)
11. Y. Deng, X. Yang, **X. Huang***, Determination of ballistic resistance model of finite thickness material based on Artificial Neural Network. *Thin-Walled Structures*, 203 (2024) 112161. (JCR Q1, IF 8.3)

Co-authored

12. H. Wei, **X. Huang**, et al., Mesoscale damage modeling of 3D woven composites under quasi-static loads: Incorporating plasticity and fiber misalignment, *Composite Structures*, 382 (2026) 120111. (JCR Q1, IF 7.8)
13. H. Wei, **X. Huang**, et al., Multiscale modeling for the impact behavior of 3D angle-interlock woven composites, *International Journal of Mechanical Sciences*, 276 (2024) 109382. (JCR Q1, IF 11.4)

14. W. Huang, W. Zhang, **X. Huang**, et al., Dynamic response of aluminium corrugated sandwich subjected to underwater impulsive loading: experiment and numerical modelling. *International Journal of Impact Engineering* 109 (2017) 78–91. (JCR Q1, IF 6.7)
15. W. Huang, B. Jia, W. Zhang, **X. Huang**, et al., Dynamic failure of clamped metallic circular plates subjected to underwater impulsive loads. *International Journal of Impact Engineering* 94 (2016) 96–108. (JCR Q1, IF 6.7)
16. Y. Deng, H. Wu, Y. Zhang, **X. Huang**, et al., Experimental and numerical study on the ballistic resistance of 6061-T651 aluminium alloy thin plates struck by different nose shapes of projectiles. *International Journal of Impact Engineering* 160 (2022) 104083. (JCR Q1, IF 6.7)
17. N. Ye, W. Zhang, D. Li, W. Huang, W. Xie, **X. Huang**, X. Jiang, Dynamic response and failure of sandwich plates with PVC foam core subjected to impulsive loading. *International Journal of Impact Engineering* 109 (2017) 121–130. (JCR Q1, IF 6.7)
18. W. Zhang, **X. Huang**, et al., Experimental investigation on ballistic stability of high-speed projectile in sand. *AIP Conference Proceedings* 1793, 060031 (2017).

INVITED TALKS & CONFERENCE PRESENTATIONS

- **Invited Speaker** — “Impact-Induced Damage Assessment and Remaining Useful Life Prediction of Lithium-Ion Cells Using Acoustic Emission Features,” PHM Korea 2026 Annual Conference, Busan, Republic of Korea (Jun 2026)
- **Oral Presentation** — “Acoustic Emission Characteristics of Lithium-Ion Cells under Mechanical Impact,” 2026 Spring Conference of the Korean Society for Noise and Vibration Engineering (KSNVE), Gangwon-do, Republic of Korea (May 2026)
- **Oral Presentation** — “Impact-Induced Pulse Power Capability Degradation in Lithium-Ion Cells: Experimental Characterization and Structural Response Analysis,” 2026 Spring Conference of the CAE and Applied Mechanics Division, Korean Society of Mechanical Engineers (KSME), Yeosu-si, Republic of Korea (May 2026)
- **Oral Presentation** — “Physics-Informed Probabilistic Machine Learning for Battery Degradation Diagnostics under Dynamic Mechanical and Electrochemical Conditions,” KSME 3rd Emerging Technologies in Mechanical Engineering (ETME 2025), Kangwon Land, Republic of Korea (Dec 2025)

STUDENT SUPERVISION

Master's Students, co-supervised with Dr. Yunfei Deng

Xiaoyue Yang 2022 – 2025

- Impact mechanics; ANN surrogate models for the impact resistance of finite-thickness plates.
- Co-authored: *Thin-Walled Structures* 203 (2024) 112161; *Thin-Walled Structures* 216 Part B (2025) 113685.
→ Now: Ph.D. Candidate, Hunan University

Han Zheng 2023 – 2026

- Mechanical safety of lithium-ion batteries: failure modes under impact and multiphysics response at the pack level.
- Co-authored: *Journal of Energy Storage* 150 (2026) 120037; *Engineering Failure Analysis* 196 (2026) 111084.
→ Now: Ph.D. Student, Sun Yat-sen University

Jiangtao Li 2023 – 2026

- Deep learning for real-time damage assessment of lithium-ion batteries under dynamic impact, with acoustic emission monitoring.
- Co-authored: *Journal of Power Sources* 662 (2026) 238737.
→ Now: Ph.D. Candidate, Beijing Institute of Technology

Yixu Lv 2023 – 2026

- Neural network surrogate models for residual velocity prediction in finite-thickness plates.
- Co-authored: *Thin-Walled Structures* 216 Part B (2025) 113685.
→ Now: Materials Engineer, AEROFUGIA

Undergraduate & Postgraduate Thesis Supervision, University of Manchester

- Instructed undergraduate student thesis: Machine learning based dynamic constitutive models (10/2023 – 04/2024)
- Instructed postgraduate (master's degree) student thesis: Signal analysis of damaged space structure by impact (08/2020 – 07/2021)
- Instructed undergraduate student thesis: Structural health monitoring of space structures under impacts (10/2020 – 03/2021)

COLLABORATION EXPERIENCE

- Physics-informed method for the prediction of dynamic flow stress data to very high strain-rate (School of Engineering, University of Manchester) (11/2024 – 06/2025)
- Multiscale modelling of 3D angle-interlock woven composites (School of Astronautics, HIT) (10/2023 – 11/2025)
- Machine learning-based lithium-ion battery damage detection and evaluation under impact (College of Aeronautical Engineering, CAUC) (11/2023 – 06/2025)

PROFESSIONAL SERVICE

- **Reviewer**, International Journal of Impact Engineering, Elsevier
- **Reviewer**, Scientific Reports, Springer Nature
- **Reviewer**, Defence Technology, Elsevier

AWARDS & GRANTS

- UK Henry Lester Trust Grant (05/2022)
- UK Henry Lester Trust Grant (05/2021)

COMPUTATIONAL SKILLS

- ABAQUS / LS-DYNA: explicit modelling of dynamic impact with user-defined subroutines (VUMAT/UMAT)
- Programming: C, Python, MATLAB, Fortran
- Deep learning: PyTorch, TensorFlow
- CATIA: geometric modelling